

$$\int \sec^n x dx = \frac{\tan x \sec^{n-2} x}{n-1} + \frac{n-2}{n-1} \int \sec^{n-2} x dx$$

$$\int \operatorname{cosec}^n x dx = \frac{-\cot x \operatorname{cosec}^{n-2} x}{n-1} + \frac{n-2}{n-1} \int \operatorname{cosec}^{n-2} x dx$$

$$Q_1. \int \cos^7 x dx$$

$$Q_2. \int \cos^5 x dx$$

$$\frac{3}{4}. \int \cos^4 x dx$$

$$\frac{4}{5}. \int \cos^8 x dx$$

For $\cot x$

$$\int \cot^5 x dx$$

$$\int \cot^7 x dx$$

For $\sec x$ and $\operatorname{cosec} x$

For $\tan x$

$$\int \tan^6 x dx$$

$$\int \tan^7 x dx$$

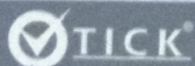
$$\int \sec^5 x dx \text{ and } \int \operatorname{cosec}^7 x dx.$$

Integrations:

$$\int \sin x dx = -\cos x + c$$

$$\int \cos x dx = \sin x + c$$

$$\int \tan x dx = \ln |\sec x| + c \text{ or } -\ln |\cos x| + c.$$



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$$\int \cot x dx = \ln |\sin x| + c$$